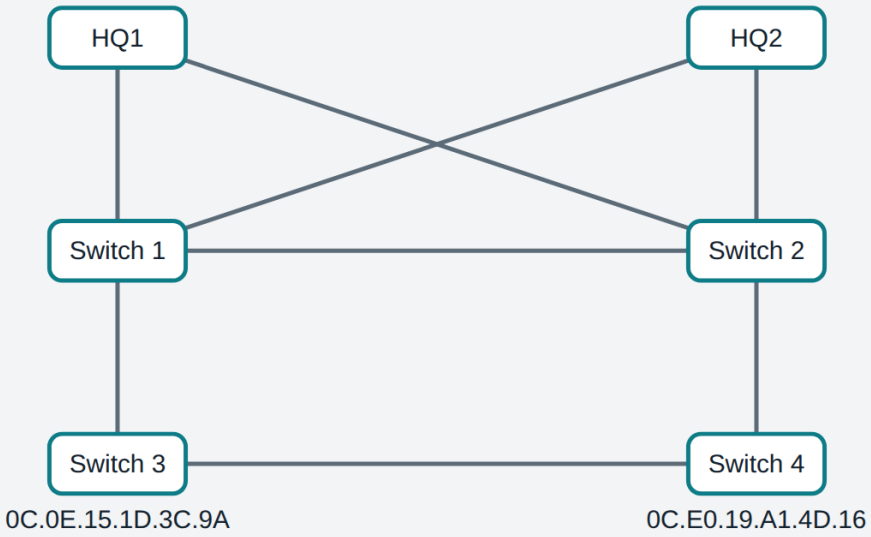
STP & VTP Questions

13 questions | Original English questions and source answers

Question 1

Refer to the exhibit. After the election process what is the root bridge in the HQ LAN?




```
Switch1: 0C.E0.38.57.24.22
Switch2: 0C.0E.15.22.1A.61
Switch3: 0C.0E.15.1D.3C.9A
Switch4: 0C.E0.19.A1.4D.16
```

Options

A. Switch 2

B. Switch 1

C. Switch 4

D. Switch 3

Answer

D. Switch 3

Source: CCNA 200-301 Questions.html | Question 1

Question 2

Which spanning-tree enhancement avoids the learning and listening states and immediately places ports in the forwarding state?

Options

A. BPDUfilter

B. Backbonefast

C. BPDUguard

D. PortFast

Answer

D. PortFast

Source: CCNA 200-301 Questions.html | Question 2

Question 3

Which result occurs when PortFast is enabled on an interface that is connected to another switch?

Options

A. VTP is allowed to propagate VLAN configuration information from switch to switch automatically.

B. Spanning tree may fail to detect a switching loop in the network that causes broadcast storms

C. Root port choice and spanning tree recalculation are accelerated when a switch link goes down

D. After spanning tree converges PortFast shuts down any port that receives BPDUs.

Answer

B. Spanning tree may fail to detect a switching loop in the network that causes broadcast storms

Source explanation

Enabling the PortFast feature causes a switch or a trunk port to enter the STP forwarding-state immediately or upon a linkup event, thus bypassing the listening and learning states.
Note: To enable portfast on a trunk port you need the trunk keyword “spanning-tree portfast trunk“

Source: CCNA 200-301 Questions.html | Question 3

Question 4

Refer to exhibit. How does SW2 interact with other switches in this VTP domain?

```
SW2
vtp domain cisco
vtp mode transparent
vtp password test
interface fastethernet0/1
description connection to SW1
switchport mode trunk
switchport trunk encapsulation dot1q
```

Options

A. It processes VTP updates from any VTP clients on the network on its access ports

B. It receives updates from all VTP servers and forwards all locally configured VLANs out all trunk ports

C. It transmits and processes VTP updates from any VTP Clients on the network on its trunk ports

D. It forwards only the VTP advertisements that it receives on its trunk ports

Answer

D. It forwards only the VTP advertisements that it receives on its trunk ports

Source explanation

The VTP mode of SW2 is transparent so it only forwards the VTP updates it receives to its trunk links without processing them.

Source: CCNA 200-301 Questions.html | Question 4

Question 5

Refer to the exhibit. What two conclusions should be made about this configuration? (Choose two)

```
SW1#show spanning-tree vlan 30

VLAN0030
Spanning tree enabled protocol rstp
Root ID  Priority  32798
         Address   0018.7363.4300
         Cost      2
         Port      13 (FastEthernet2/1)
         Hello Time 2 sec  Max Age 20 sec  Forward Delay 20 sec
<output omitted>
```

Options

A. The spanning-tree mode is PVST+

B. The spanning-tree mode is Rapid PVST+

C. The root port is FastEthernet 2/1

D. This is a root bridge

E. The designated port is FastEthernet 2/1

Answer

B. The spanning-tree mode is Rapid PVST+

C. The root port is FastEthernet 2/1

Source explanation

This bridge is not the root bridge because it does not have the statement “This bridge is the root”. When the local switch is not the root bridge, the port it shows would be the root port to the root bridge. Therefore in this case FastEthernet2/1 is the root port that is connected to the root bridge.

Source: CCNA 200-301 Questions.html | Question 5

Question 6

How does STP prevent forwarding loops at OSI Layer 2?

Options

A. MAC address forwarding

B. Port blocking

C. Collision avoidance.

D. TTL

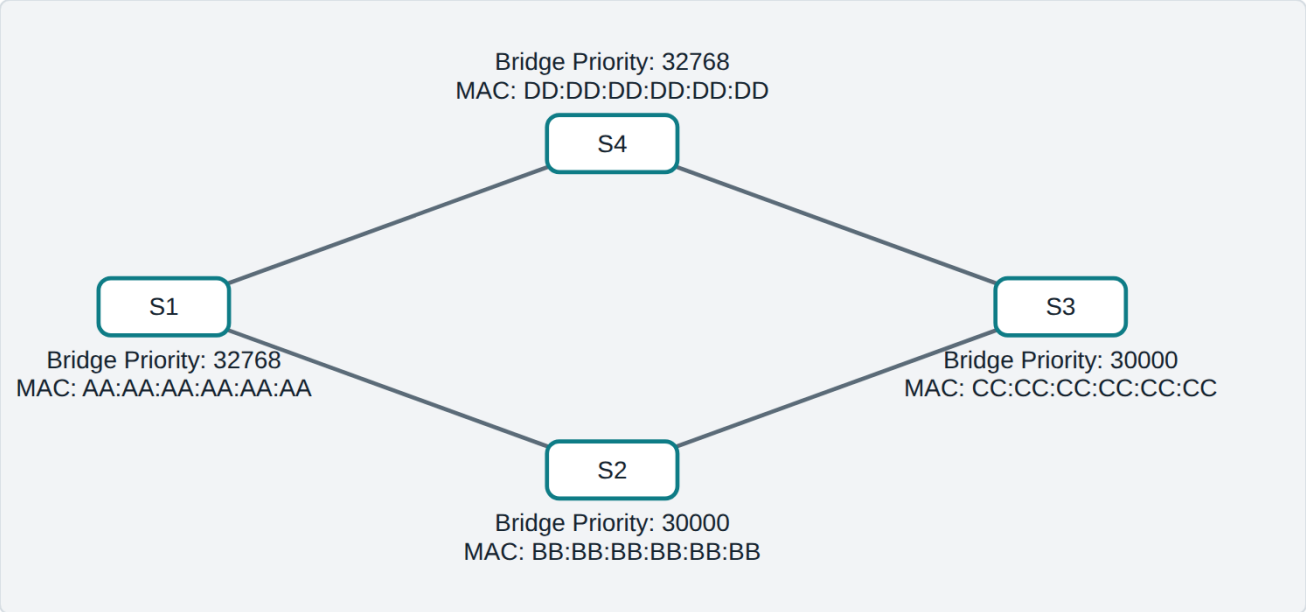
Answer

B. Port blocking

Source: CCNA 200-301 Questions.html | Question 6

Question 7

Refer to the exhibit. Which switch becomes the root bridge?



Options

A. S3

B. S1

C. S2

D. S4

Answer

C. S2

Source: CCNA 200-301 Questions.html | Question 7

Question 8

Which state does the switch port move to when PortFast is enabled?

Options

A. learning

B. listening

C. forwarding

D. blocking

Answer

C. forwarding

Source: CCNA 200-301 Questions.html | Question 8

Question 9

Refer to the exhibit. Which switch in this configuration will be elected as the root bridge?




```
SW1: 0C:E0:38:00:36:75
SW2: 0C:0E:15:22:05:97
SW3: 0C:0E:15:1A:3C:9D
SW4: 0C:E0:18:A1:B3:19
```

Options

A. SW4

B. SW2

C. SW1

D. SW3

Answer

D. SW3

Source explanation

As this question did not state about bridge priority so we can assume all of them are using the default priority. Therefore the switch with lowest MAC address wins the election.

Source: CCNA 200-301 Questions.html | Question 9

Question 10

What criteria is used first during the root port selection process?

Options

A. lowest neighbor’s bridge ID

B. lowest path cost to the root bridge

C. lowest neighbor’s port ID

D. local port ID

Answer

B. lowest path cost to the root bridge

Source: CCNA 200-301 Questions.html | Question 10

Question 11

When using Rapid PVST+, which command guarantees the switch is always the root bridge for VLAN 200?

Options

A. spanning-tree vlan 200 root primary

B. spanning-tree vlan 200 priority 38572422

C. spanning-tree vlan 200 priority 614440

D. spanning-tree vlan 200 priority 0

Answer

D. spanning-tree vlan 200 priority 0

Source: CCNA 200-301 Questions.html | Question 11

Question 12

What is the primary effect of the spanning-tree portfast command?

Options

A. It enables BPDU messages

B. It immediately puts the port into the forwarding state when the switch is reloaded

C. It minimizes spanning-tree convergence time

D. It immediately enables the port in the listening state

Answer

C. It minimizes spanning-tree convergence time

Source explanation

The purpose of Port Fast is to minimize the time interfaces must wait for spanning-tree to converge, it is effective only when used on interfaces connected to end stations.

Source: CCNA 200-301 Questions.html | Question 12

Question 13

Which configuration ensures that the switch is always the root for VLAN 750?

Options

A. `Switch(config)#spanning-tree vlan 750 priority 38003685`

B. `Switch(config)#spanning-tree vlan 750 priority 0`

C. `Switch(config)#spanning-tree vlan 750 root primary`

D. `Switch(config)#spanning-tree vlan 750 priority 614440`

Answer

B. Switch(config)#spanning-tree vlan 750 priority 0

Source explanation

The keyword here is "always" so we should set the priority to the lowest value which is 0. If we use the "primary root" keyword then our switch is only set to the lower value compared to other switches. And new switch added in the future can be set to lower value than our switch to take over the root role.

Source: CCNA 200-301 Questions.html | Question 13